Week 11 Problem Set
Markov Asset Pricing

Dynamic Programming and Quantitative Macroeconomics

Required reading: `https://python.quantecon.org/markov_asset.html`

Instructions. Submit your completed Jupyter notebook with derivations in Markdown cells, executable Python code, figures, and brief economic interpretations. All six problems receive equal weight. No external data files are needed. Use numerical checks where requested.

Baseline Economy

Let $\beta = 0.95$, $\gamma = 2$, and let next-period gross consumption and dividend growth depend on next-period Markov state j:

$$g = (0.98, 1.02, 1.06), \qquad P = \begin{pmatrix} 0.80 & 0.15 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.15 & 0.80 \end{pmatrix}.$$

States are recession, normal, and expansion. P_{ij} is the probability of state j tomorrow conditional on state i today. Assume CRRA preferences, $u'(c) = c^{-\gamma}$.

Problem 1: 1. Markov Growth Economy

(a) Verify that P is a stochastic matrix.
(b) Compute the invariant distribution π satisfying $\pi P = \pi$ and $\sum_i \pi_i = 1$.
(c) Simulate 10,000 states, beginning in the normal state, with a fixed random seed. Compare the empirical frequencies to π.
(d) Interpret persistence and the long-run mean gross growth factor $\sum_i \pi_i g_i$.

Problem 2: 2. Euler Equation and Stochastic Discount Factor

(a) Starting from $u'(c_t) = \beta \mathbb{E}_t[R_{t+1} u'(c_{t+1})]$, derive $1 = \mathbb{E}_t[m_{t+1} R_{t+1}]$ with $m_{t+1} = \beta u'(c_{t+1})/u'(c_t)$.
(b) Under CRRA utility and $c_{t+1}/c_t = g_j$, derive $m_{ij} = \beta g_j^{-\gamma}$.
(c) Compute and display the three possible values of the SDF. Explain why low-growth outcomes have larger discount factors.

Problem 3: 3. Lucas Tree and Matrix Pricing

Assume aggregate consumption equals dividends and $d_{t+1} = g_j d_t$ when next period's state is j.
(a) Starting from $p_t = \mathbb{E}_t[m_{t+1}(p_{t+1} + d_{t+1})]$, show that the state-dependent price-dividend ratio satisfies $v_i = \beta \sum_j P_{ij} g_j^{1-\gamma}(1 + v_j)$.

(b) Define $J_{ij} = P_{ij} g_j^{1-\gamma}$ and derive $(I - \beta J)v = \beta J \mathbf{1}$. State the condition ensuring a finite solution.
(c) Solve with `np.linalg.solve` (not explicit matrix inversion).
(d) Starting from zero, iterate $v^{(k+1)} = \beta J(\mathbf{1} + v^{(k)})$ until the sup-norm difference is below 10^{-11}. Compare results and verify the residual.

Problem 4: 4. Risk Aversion and Persistence

(a) For $\gamma \in \{0, 1, 2, 4, 6, 10\}$, compute the three price-dividend ratios and graph them against γ.
(b) Interpret the cross-state and risk-aversion patterns; avoid claiming prices must fall with γ in every economy.
(c) Form $P(\lambda) = \lambda P + (1 - \lambda)\mathbf{1}\pi$, where π is the stationary distribution from Problem 1 and $\lambda \in [0, 1]$. Verify that every $P(\lambda)$ has invariant distribution π. Plot price-dividend ratios versus λ at $\gamma = 2$ and interpret the effect of persistence.

Problem 5: 5. Risk-Free Bonds and Consols

(a) A one-period risk-free bond pays one unit next period. Show that its price in state i is $q_i = \beta \sum_j P_{ij} g_j^{-\gamma}$ and its net yield is $r_{f,i} = q_i^{-1} - 1$. Compute both.
(b) A consol pays one unit of the consumption good each period starting next period. Show that $b_i = \beta \sum_j P_{ij} g_j^{-\gamma}(1 + b_j)$. Define $H_{ij} = P_{ij} g_j^{-\gamma}$ and solve the linear system.
(c) Verify both pricing equations numerically and explain the distinction between the consol price and the Lucas-tree price-dividend ratio.

Problem 6: 6. Log Utility and Synthesis

(a) Prove analytically that when $\gamma = 1$, $v_i = \beta/(1 - \beta)$ for every state, provided the relevant transversality condition holds.
(b) Verify numerically and report the maximum absolute error.
(c) Show that the Lucas-tree pricing matrix is $J = P$ under log utility.
(d) Explain in a paragraph how the Euler equation, Markov transition matrix, and linear algebra fit together to determine asset prices.
(e) Optional: explain what changes if dividends are not identical to aggregate consumption.